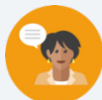


6.11 Solving Problems Using Similarity – Part 1

Lesson Introduction

 Benchmark	MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.		 Learning Target	I can solve mathematical problems involving similarity in two-dimensional figures.
 Benchmark Coherence	Building On MA.8.GR.2.4 Solve mathematical and real-world problems involving proportional relationships between similar triangles.		Working Toward N/A	
 Essential Questions	- How are similar figures used to find missing values? - How do you use scale factor to determine the length of a side?			
 Lesson Narrative	This lesson extends the previous lesson by using the relationships between angles and sides of similar figures to create equations. These equations are then solved to find the values of variables and missing angle and side measures. Student knowledge of scale factor is key in determining and writing the correct equation.			
 Component Summary	6.11.1 Warm-Up (~5 min.)	Applying similar triangles to model a building (<i>SE p. 354</i>)	6.11.3 Your Turn! (10-15 min.)	Determining measures using similar figures (<i>SE p. 356</i>)
	6.11.2 Collaboration (20-25 min.)	Determining measures using similar figures and error analysis (<i>SE p. 354</i>)	6.11.4 Wrap-Up (~5 min.)	Analyzing errors in sample work (<i>SE p. 358</i>)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Similarity - Congruent - Proportional - Corresponding		(Optional) Calculators	If using “area experts” for 6.11.3 Your Turn!, create question cards for the expert to use while helping students.

 Teacher Tips	Common Misconceptions - Students may set up equations incorrectly by setting the wrong angles equal to each other. - Students may multiply by the wrong scale factor when determining lengths.	Things to Consider Consider utilizing “area experts” for 6.11.3 Your Turn!. While monitoring 6.11.2 Collaboration, listen and look for students who demonstrate understanding of the lesson learning targets. Assign each as the expert for one of the questions 1a through 1e. If other students have questions, they seek out the area expert as the first resource. The area expert should not give out answers, but instead assist students by looking at their work and asking guiding questions.
	Literacy and Language Routines Stronger and Clearer Each Time As students progress through this lesson, they may find opportunities, particularly during 6.11.2 Collaboration, to use the Stronger and Clearer Each Time routine to encourage students to develop responses that are stronger (often longer) with better justifications and clearer with more precise terms and linked, organized, complete sentences.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR1.1: Participate in Effortful Learning 6.11.3 Your Turn! offers students an opportunity to embrace the concept of effortful learning. Students can solve equations on their own or in small groups or even help others as they monitor peers for success.
	 Differentiate Instruction	
Lesson Notes:		

6.11.1 Warm-Up: The Flatiron Building

Component Overview: Students consider how similar figures might be used in the real world.



6.11.1 Warm-Up: The Flatiron Building

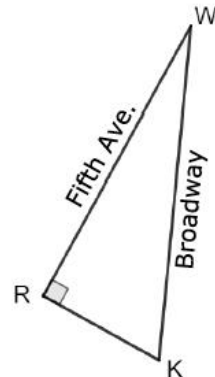


The Flatiron Building, designed by Daniel Burnham and built in 1902, is a triangular 22-story building located in New York City. The building sits on a triangular block formed by 5th Avenue, Broadway, and East 22nd Street. The building got its name from an old-fashioned flat iron.



In what industries or specific situations might similar figures be useful?

Although the ground base appears to be an isosceles triangle, the base of the Flatiron Building is actually a right triangle. Morgan used a map to draw a triangle that models the base of the building.



1. What other measurement should Morgan check to be sure her drawing is similar to the building?

Answers will vary. Morgan should check $\angle RWK$ or $\angle WKR$ for congruence.

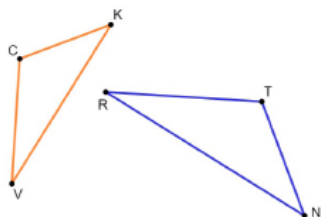
6.11.2 Collaboration: Similar Figures

Component Overview: Students will use the relationship between the similar figures to create equations needed to solve for unknown variables, angles, and side lengths. Close attention should be paid to what the scale factor represents and how it is used in the equations.



6.11.2 Collaboration: Similar Figures

1. $\triangle CKV$ and $\triangle TNR$ are shown, where $\triangle CKV \sim \triangle TNR$.



- a. Some measurements of the angles in $\triangle CKV$ and $\triangle TNR$ are given.
 $m\angle TRN = 28^\circ$, $m\angle VCK = (8x + 26)^\circ$, and
 $m\angle RTN = (12x - 8)^\circ$.

Determine each.

$$x = \underline{8.5} \quad m\angle CKV = \underline{58^\circ}$$

Some measurements of the sides of $\triangle CKV$ and $\triangle TNR$ are given.

$$VK = 8, CK = 5z + 1.7, CV = 5.4, TR = 2y - 1.25, \\ RN = 10, \text{ and } TN = 5.25.$$

- b. What is the scale factor from $\triangle CKV$ to $\triangle TNR$?

$$\frac{5}{4}$$

- c. Determine the value of each variable.

$$z = \underline{0.5} \quad y = \underline{4}$$

- d. Determine the measure of each side.

$$CK = \underline{4.2} \quad TR = \underline{6.75}$$



Common Misconception

If students found the value of x to be 0.25 or 3, then they set the wrong angles equal to each other in the equation.

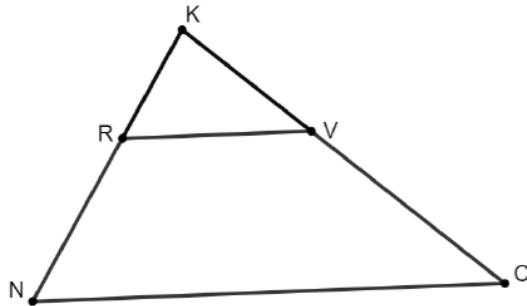


Consider prompting struggling students, if needed.

- Which triangle is the larger of the two?
- Is a side length of triangle CKV larger or smaller than the corresponding side length of triangle TNR ?
- Should the scale factor that will be multiplied by the sides of triangle CKV be larger than 1 or between 0 and 1 to find the corresponding side length of TNR ?

6.11.2 Collaboration: Similar Figures (continued)

2. $\triangle KVR$ and $\triangle KCN$ are shown, where $\triangle KVR \sim \triangle KCN$.



Some measurements of the sides are given.

$RV = 4.6$, $KR = 3$, $KV = 5x - 11$, $NC = 11.5$, and

$KC = 3x + 1$.

Maria used the information to determine the length of $\overline{KC}$. Her work is shown.

Maria's Work	
First, Maria found the scale factor.	$\frac{NC}{RV} = \frac{11.5}{4.6} = 2.5$
Second, she used the scale factor to write an equation. Then she solved for x .	$5x - 11 = 2.5(3x + 1)$ $5x - 11 = 7.5x + 2.5$ $-2.5x = 13.5$ $x = -5.4$
Lastly, she substituted x into an equation.	$KC = 3(-5.4) + 1$ $KC = -15.2$



Students may struggle differentiating the two triangles. Consider having students draw two separate triangles labeling the smaller one triangle KRV and the larger one triangle KNC .

6.11.2 Collaboration: Similar Figures (continued)

Maria knows side lengths cannot be negative values, but she does not know what she did wrong.

- Find and circle Maria's error.
- Write a note to Maria explaining what her mistake was and how she can learn from the mistake.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

Answers will vary.



Consider helping students find the error by asking the following probing questions, as needed.

- Does this scale factor represent a reduction or an enlargement?*
- Which triangle side lengths will you multiply this scale factor by to determine the lengths of the corresponding side lengths in the similar triangle?*

6.11.3 Your Turn!

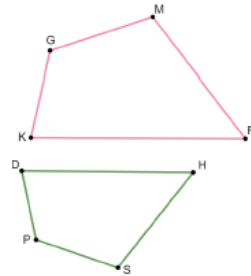
Component Overview: Students will practice finding unknown values in similar figures.



6.11.3 Your Turn!

1. $GMRK$ and $PSHD$ are shown, where $GMRK \sim PSHD$.

Some measurements of the sides of $GMRK$ and $PSHD$ are given.



$KG = 12$, $GM = 4x - 12$, $MR = 7k - 15$, $DP = 9.6$, $SP = 12.8$, $KR = 30$, and $SH = 4k - 4$.

- a. Determine the value of each variable.

$$x = \underline{7} \quad k = \underline{5}$$

- b. Determine the value of each side.

$$GM = \underline{16} \quad DH = \underline{24}$$

$$SH = \underline{16}$$

- c. What is the perimeter of $GMRK$?

$$\underline{78}$$

Some measurements of the angles of $GMRK$ and $PSHD$ are given.

$$m\angle KGM = (11w - 12)^\circ, m\angle GMR = 110^\circ,$$

$$m\angle GKR = (6z + 18)^\circ, m\angle DPS = (8w + 24)^\circ, \text{ and}$$

$$m\angle PDH = (8z - 2)^\circ.$$

- d. Determine the value of each variable.

$$w = \underline{12} \quad z = \underline{10}$$

- e. Determine the value of each angle.

$$m\angle KGM = \underline{120^\circ} \quad m\angle DHS = \underline{52^\circ}$$

$$m\angle PDH = \underline{78^\circ}$$



Consider utilizing “area experts” for each of the questions 1a through 1e. If other students have questions, they seek out the area expert as the first resource. The area expert should not give out answers but instead assist students by looking at the other students’ work and asking guiding questions.

Consider putting questions the expert might ask on cards to give to each expert. Consider having two experts for 1a, as more questions may arise there.

- 1a – What is the scale factor? If your scale factor is greater than 1, did you multiply it by the side length of the smaller figure? If your scale factor is between 0 and 1, did you multiply it by the side length of the larger figure? What steps did you use to solve the equation?
- 1b – What value did you substitute in for the variable? Did you follow the order of operations?
- 1c – What is the definition of perimeter? What are the lengths of each side?
- 1d – Why did you set those expressions equal? What steps did you use to solve the equation?
- 1e – What value did you substitute in for the variable? Did you follow the order of operations?

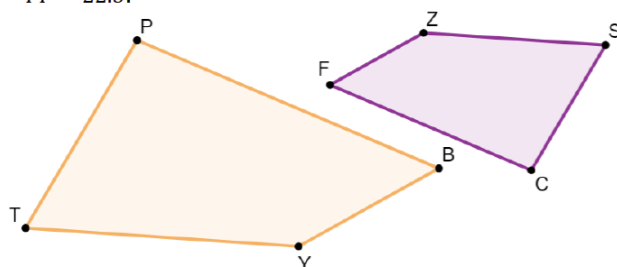
6.11.4 Wrap-Up: Error Analysis

Component Overview: Students will analyze two work samples and determine if either are correct. Students must justify their responses. Then students find the length of a side of one of the similar triangles. Students should be allowed to access a calculator to complete 6.11.4 Wrap-Up.



6.11.4 Wrap-Up: Error Analysis

- Cheryl and Austin are finding the measure of side SC . They were given the following information about similar quadrilaterals $ZSCF$ and $YTPB$ that are shown. $ZSCF \sim YTPB$, $ZS = 15$, $SC = 5x - 8$, $PT = 3x + 6$, and $YT = 22.5$.



Cheryl's and Austin's equations are shown.

Cheryl's Work	Austin's Work
$15(5x - 8) = 22.5(3x + 6)$	$1.5(5x - 8) = 3x + 6$

- Determine whose equation, if any, is set up incorrectly. Show your work or write an explanation.
Answers may vary. Cheryl's equation is incorrect. Cheryl should have written the equation $22.5(5x - 8) = 15(3x + 6)$ because the proportion relating the side lengths of the quadrilaterals could have been written $\frac{22.5}{15} = \frac{3x+6}{5x-8}$.
- What is the length of $\overline{SC}$?
 $SC = 12$



Use the responses in this activity to determine any remediation needs.

- If students state Cheryl's work is correct, they may need help on setting up a proportion.
- If students state Austin is correct, they may need help understanding what a scale factor is and how it is used.
- If students miss question 1b, they may have set up or used the wrong equation or they may need help solving multistep equations.